Next-generation brand protection goes beyond serialization

AuthentiQ's **CLOUD-BASED, CONSUMER-FRIENDLY ARTIFICIAL INTELLIGENCE-DRIVEN TECHNOLOGY** adds functionality to images, alphanumerics, QR codes, barcodes, RFID and NFC chips.

Rick Lingle, Technical Editor

While serialization is a big first step toward helping boost a product's security and trackability through the supply chain, it also creates potential loopholes. What if there were a way to close the loopholes through anti-counterfeiting authentication while also providing marketplace traceability and contextual marketing, all through the use of conventional mobile devices?

There is now with the introduction of the patent-pending **authentiQ** (http://au.thentiQ.com) technology that relies on the Cloud and Artificial Intelligence programming to help make that happen.

CEO Adéna Shao responds to *Packaging Digest*'s questions about this robust and intriguing new multilayered tool for brand owners and others.

Easy for consumers to use, authentiQ's AI-driven software provides a quantified high rate of authenticity.

What is authentiQ?

Shao: At its core, authentiQ's technology detects copies of your products in the marketplace. On the back of this technology, we also bring marketplace intelligence data to you. Imagine you're a hot brand, but leaking profit to counterfeits. As a brand manager, would you like to know that you've recovered true marketshare each time a copy is detected in the marketplace? Not to mention who is interested in your product and where? As a brand defense director, would you like to report to management in concrete terms the effectiveness of the solution you chose?

How did it come about?

Shao: Like others, we started with serialized codes at the salable item level in tackling product authentication. We still start with serial codes, either codes that are generated by us, or those that clients already have.

However, it's no secret that serial codes by themselves do not protect from copies. For example, multiple copies can be made of an existing code. How do you know which one is likely to be the original code or a copy? Another concern well known among industry insiders is the low rate of codes being checked, which brings into question the effectiveness of serialized codes in the first place. authentiQ offers a solution that addresses both these concerns.

How does it work?

Shao: The codes we generate can be open (overt) codes on the packaging or covert, or a combination of multiple overt and covert codes. The consumer can check the overt code prior to purchase or post purchase via his mobile device by scanning, near-field communication (NFC), texting or a phone call. The codes can also be checked by distribution channel staff and by law enforcement. We find overt codes superior to covert ones in consumer friendliness, since the former encourage scanning/checking by the general public.

Each code checked is run through our artificial intelligence software to determine its plausibility of being your real product or not. I am unable to disclose details, but let it be known that even at a scan rate of only 0.66%, our fraud detection mechanism is quantified at 99.995+% by Gunnar Pruessner, Ph.D., lecturer in Mathematical Physics and distinguished author. Pruessner's publications include *Measuring Probability Densities in Complex Systems,* and *Self-Organised Criticality: Theory, Models and Characterisation* (Cambridge University Press).

What does it do better than other methods?

Shao: The salient point is we've taken the next step to detect copies, tackling the gaping problem of weeding out false positives and false negatives of overt codes. This is a seismic advantage above other systems that simply tell you how many times your code has been checked. Our goal is to furnish information that is as accurate as possible regarding copies detected. Here I quote Dr. Pruessner, "Compelling and impressive. As I quantified authentiQ's solution, it surprised even me to find that authentiQ can achieve such a high copy detection rate with very low scan rates. 99.995+% copy detection with just a 0.66% scan rate is impressive."

While fraud detection can be achieved with very low check rates, a brand can still directly influence scan rates using our marketing features: coupon/loyalty programs are easily controlled through our system. To be clear, we are not providers of QR codes. Rather, we provide useful functionality behind the codes, whatever form the codes might take on a brand's packaging: images, alphanumerics, quick response codes (QRs), barcodes, radio frequency identification (RFID). We give a higher level of confidence.

Please provide examples of how this works.
Shao: Imagine a new mother in Xiamen, a growing Tier 2 city of more than 3.5 million residents in China, shopping online for imported milk powder. She can scan a code during delivery to verify the provenance of your product.

Or imagine a distributor in India who receives a shipment of your luxury spirits. He can scan codes we provide to you during his receiving process to verify the provenance of that delivery.

Yet again, imagine a cognac connoisseur in China who enjoys your products. While he scans a code to accumulate loyalty points, we check for product authenticity and he (and you) are alerted should there be an anomaly.

Or imagine a diabetes patient in the United States or Europe monitoring her blood sugar with your test strips and consumer app. We can provide integration with your app, so that the patient is prompted and able to verify the authenticity of her new pack of test strips, all during the normal course of her diabetes treatment.

What are the steps to implementation?
Shao: Our abiding principle is ease of use for each stakeholder, so the brand should incur zero internal IT costs, and zero friction on the production line.

Steps to implementation are easy:
1. The client simply signs on to our platform.
2. After sharing the relevant information with our system, either through our web front-end, or through ERP integration, we generate the serial codes, which the client applies on their packaging off the production line.

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— Adena Shao, CEO, authentiQ

To preserve production line speeds, we offer brands several options on how to apply the codes:
• using track and trace machines,
• pre-printed on packaging before filling, or
• printed on separate labels. Labels and RFIDs can be applied on the production line.

Our pricing is competitive. We guarantee the value you gain as a brand is measurable in hard dollars.

What options are available to customers?
Shao: The brand has wide discretion on how the code looks on their packaging. We are well aware that branding does not and should not take a secondary seat to technology. We can integrate images, alphanumerics, QRs, bar codes, RFID chips. When mobile devices become equipped with yet another way to connect with physical goods, we'll integrate that capability as well.

Certain forms of the codes are more relevant for certain sectors and markets. Where smartphone penetration is high, as it is in China, QR codes have an advantage in that consumers are used to scanning them.

In Africa, consumers tend to text more. Checking post- versus pre-purchase can also factor in to how codes manifest on packaging. In addition, a brand's consumer engagement approach differs from market to market. We therefore also provide the brand deep marketing capabilities if they wish to attach marketing to the codes; for example, running coupon, loyalty programs and contextual marketing, the latter of which is another topic in and of itself.

Our philosophy is that technology must be useful to those who use it; hence, we offer any necessary flexibility.

What is the current status?
Shao: Watch for our partnership with a super-premium spirits brand in 2015. We're also in advanced discussions with early adopters, who are recognized innovators themselves. and we are keen to roll out our solution with leaders in other high-value branded categories. PD